

ZEHUA ZHANG

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EDUCATION

ARIZONA STATE UNIVERSITY

Tempe, AZ

PhD Student in Computer Science | GPA: 3.67 / 4.0

Aug 2024 – Present

Co-advised by Dr. Chitta Baral and Dr. Ruoyu Wang

Expected completion: Spring 2028

CARNEGIE MELLON UNIVERSITY

Pittsburgh, PA

M.S. in AI Engineering – Information Security | GPA: 3.7 / 4.0

May 2024

Advised by Dr. Amir Barati Farimani and Dr. David Varodayan

BOSTON COLLEGE

Chestnut Hill, MA

B.A. in Computer Science and Economics | GPA: 3.65 / 4.0

May 2022

Dean's List (First Honor): 2020-2022; Highlighted Schiller Student Challenge Projects

RESEARCH EXPERIENCE

ARIZONA STATE UNIVERSITY

Tempe, AZ

Graduate Research Assistant

Aug 2024 – Present

Constructed benchmark dataset for evaluating agentic open-source software compilation methods, enabling downstream research in software engineering and binary security.

Designed a new multi-agent compilation method enhanced with LLM-assisted Retrieval module that surpasses the strongest rule-based baseline by approximately 50%.

Proposed a jailbreaking technique of encoding malicious queries with layered novel ciphers that exploits LLM's ability of interpreting complex instructions and decode complex user-defined ciphers, increasing jailbreaking success rates from 40% to 78%.

CARNIGIE MELLON UNIVERSITY

Pittsburgh, PA

Graduate Research Assistant

May 2023 – May 2024

Utilized Deep Learning methods to surrogate molecular dynamics; proposed a new masking pretraining method for Force and Energy-Centric Graph Neural Networks.

Pretrained and finetuned GNNs to predict atomic force and potential energy of water molecule datasets (Tip3p and RPBE) and small organic molecule dataset (MD17), reducing respective RMSE for water datasets by 38.09% and 25.4%.

PEKING UNIVERSITY
Research Assistant

Beijing, China
May 2021 – June 2021

Evaluated and tested multiple protocols of Blockchains and Smart Contract codes, reporting on best practices of Smart Contract Development.

Constructed Test net based on Morden and Private blockchain to reproduce 4 attacks targeting Smart Contracts and qualitatively analyzed the significance of Governance Token on restricting Flash Loan Attack.

Quantitatively assessed suppressing effecting of regulations on Cryptocurrency transactions and major mining pools based on on-chain data.

RESEARCH INTERESTS

General Interests: Natural Language Processing and its applications on Security/SE
Recent Focuses: LLM Post-training, Multi-Agent System, AI for Software Security

PROFESSIONAL EXPERIENCE

SAMSUNG RESEARCH AMERICA
Research Intern

Mountain View, CA
May 2026 – Aug 2026

Conducted research on indirect prompt injection (IPI) vulnerabilities in LLM-based agentic systems, with a focus on improving robustness against adversarial instructions embedded in untrusted data.

Developed a gym-like training environment for scalable IPI training data sampling and generation, supporting both online and offline post-training of assistant models and standalone safety verification models.

Trained LLMs of different sizes and families using SFT and RL (DPO and GRPO) and achieved state-of-the-art performance on established benchmarks, including AgentDojo and AgentDyn.

SOHU.INC
Algorithm Development Intern

Beijing, China
June 2021 – Sept 2021

Designed and deployed a real-time feed deduplication system to enhance user experience and content diversity for social media app.

Leveraged large-scale data streaming pipelines (Kafka, Spark) and advanced similarity detection methods (e.g., simhash, Bloom filters) to filter redundant multi-media data.

Achieved substantial efficiency gains, reducing repetitive text exposure by ~90% and duplicate video content by ~30%.

TECHNICAL SKILLS

Programming: Python, C, Scala, Java, R, Swift, JavaScript, HTML, CSS

ML/DL Frameworks: PyTorch, TensorFlow, PyG, Verl, vLLM

Data & Distributed Systems: Spark, Kafka, Kubernetes

Cloud & Databases: MySQL, MongoDB, Hive, HBase; GCP, AWS

Software Development: React, IOS, Docker, MCP

PUBLICATIONS

A. Saeidi, **Z. Zhang**, R. K. Singh, et al., “CAST: Critique-Aware Supervision for Training Reliable Long-Horizon Tool-Calling Agents.”
<https://arxiv.org/abs/2608.30147v1>. Accepted to EMNLP 2026 Main Conference. (*Co-First Author*)

Z. Zhang, A. Bajaj, D. Handa, et al., “Build-Bench: Benchmarking LLM Agents on Compiling Real-World Open-Source Software.” <https://www.arxiv.org/abs/2509.25248>.
Accepted in TMLR. (*First Author*)

D. Handa, **Z. Zhang**, A. Saeidi, S. Kumbhar, and C. Baral, “When ‘Competency’ in reasoning opens the door to vulnerability: jailbreaking LLMs via novel complex ciphers,”
arXiv.org, Feb. 16, 2024. <https://arxiv.org/abs/2402.10601>. Accepted for presentation at NeurIPS 2025 Reliable ML from Unreliable Data workshop.

Z. Zhang, Z. Li, and A. B. Farimani, “Masked pretraining strategy for neural potentials,”
AIP Advances, vol. 14, no. 9, Sep. 2024, doi: 10.1063/5.0202647. (*First Author*)

PREPRINTS

One paper submitted to NDSS 2027. (*First Author*)

J. Li, Y. Yuan, and **Z. Zhang**, “Enhancing LLM Factual Accuracy with RAG to Counter Hallucinations: A Case Study on Domain-Specific Queries in Private Knowledge-Bases,”
arXiv.org, Mar. 15, 2024. <https://arxiv.org/abs/2403.10446> (*Co-First Author*)

TEACHING EXPERIENCE

ARIZONA STATE UNIVERSITY
Graduate Teaching Assistant

Tempe, AZ
Aug 2026 – Dec 2026

Courses: Information Assurance

Responsibilities: grading, office hours, student support

BOSTON COLLEGE
Undergraduate Teaching Assistant

Chestnut Hill, MA
Sept 2021 – May 2022

Courses: Randomness and Computation (CSCI2244), Web Application Development (CSCI2254)

Responsibilities: grading, office hours, student support